

"Don't Say It!": Constraints, Compliance, and Communication when Language Models Play Taboo

Sara Candussio^{1,†}, Francesca Padovani^{2,†}, Daniel Scalena^{2,3,†} and Malvina Nissim²

¹*ALLab, MIGe, Univeristy of Trieste, Italy*

²*Center for Language and Cognition (CLCG), University of Groningen, The Netherlands*

³*University of Milano - Bicocca, Italy*

Abstract

The game of Taboo requires describing a target word without using a set of forbidden words, so that other players can guess it. This deceptively simple task combines strict lexical constraints with the need for communicatively effective descriptions, making it a compelling playground for examining how LLMs navigate competing demands at inference time. We evaluate two open-weight models under conditions that intervene at progressively deeper levels of the generative process, from prompting to generation-time constraints to internal representations manipulations. We assess their outputs through forbidden word violation detection, LLM-as-a-judge measuring the degree to which generated descriptions successfully evoke the target concept for both human and machine guessers, and examining whether the strategies models adopt under constraint align with those of human players. Our results show that compliance with the rules of the game and communicative effectiveness trade off differently across conditions, and that models remain substantially weaker than humans as guessers, suggesting that lexical grounding under constraint is an open challenge for current language models¹.

Keywords

taboo, prompting, constrained generation, word-guessing, SAEs

1. Introduction

The game of Taboo requires a player to describe a target word without using a set of forbidden words, so that another player can guess it. Beyond its appeal as a *parlour* game, Taboo instantiates a linguistically interesting challenge: the speaker must suppress some lexically and conceptually salient words while simultaneously producing a description that is informative enough to identify the target. This combination of constraint satisfaction and communicative effectiveness makes it a particularly suitable setting for probing language models in a setting that goes beyond standard instruction-following benchmarks. In particular, it remains unclear whether descriptions generated under constraint conform to the demands of the task — namely whether

¹Code and data can be found at <https://github.com/DanielSc4/LMtaboo/>

CLiC-it 2026: Twelfth Italian Conference on Computational Linguistics, September 14 — 16, 2026, Palermo, Italy

*Corresponding author.

[†]These authors contributed equally.

✉ sara.candussio@phd.units.it (S. Candussio); f.padovani@rug.nl (F. Padovani); d.scalena@rug.nl (D. Scalena); m.nissim@rug.nl (M. Nissim)

🆔 0009-0004-5198-6970 (S. Candussio); 0009-0007-3489-9631 (F. Padovani); 0009-0006-0518-6504 (D. Scalena); 0000-0001-5289-0971 (M. Nissim)



© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

they are sufficiently detailed, salient enough to evoke the target concept, and communicatively effective enough for the game to be successfully completed.

In this paper we investigate how LLMs play Taboo in Italian, evaluating two open-weight models under conditions that intervene at progressively deeper levels of the generative process: from prompting, to generation-time constraints, to the manipulation of internal representations. To ground our evaluation in an actual game-play, we conduct a human study in which the roles are reversed in both directions: human evaluators read model-generated descriptions and attempt to guess the target word, and vice versa. This bidirectional design allows us to directly assess how well descriptions produced succeed in conveying the target concept, and whether the descriptive choices adopted by models and humans are mutually interpretable.

Our findings reveal that the relationship between rule compliance and description quality varies substantially across conditions, and that models fall considerably short of human performance in the guessing role, pointing to lexical grounding under constraint as an open challenge for current language models.

2. Related works

Language games as NLP benchmarks Word-based games have been exploited as playgrounds for NLP, offering well-defined rules and measurable objectives suitable for benchmarking. Within the Italian NLP community, language games have attracted growing attention as probes for linguistic competence: the TV game *La Ghigliottina* has been tackled both with dedicated NLP systems [1] and used to benchmark LLMs [2], while studies on rebus and cross-word solving reveal consistent limitations in tasks requiring compositional, phonological, and lateral reasoning [3, 4, 5, 6]. Beyond Italian, benchmarks such as NYT Connections [7] and Codenames [8] consistently show a substantial gap between LLM and human performance on tasks requiring deliberate reasoning and theory of mind. Word games have also been studied as training environments: Horst et al. [9] show that LLMs can learn from dialogue game self-play feedback, with Taboo among the training games. Closest to our work, Cywiński et al. [10] study Taboo specifically in the context of LLM interpretability, using the game as a test for eliciting latent knowledge. Our work differs in scope: rather than focusing on a single intervention, we systematically compare constraint enforcement strategies at multiple levels of the generative process and we furthermore ground the evaluation in an actual human game-play.

Constrained text generation Comprehensive evaluations of constrained text generation across open-source LLMs show that models exhibit systematic deficiencies even on relatively simple lexical constraints. Prompt-based approaches suffer from position bias and struggle with morphologically complex forms, and stricter format constraints have been demonstrated to cause performance degradation [11, 12]. Ciaccio et al. [13] evaluate several Italian LLMs on their ability to generate sentences adhering to morpho-syntactic specifications, finding systematic deficiencies across models and constraint types. Calderaro et al. [14] further show that even state-of-the-art proprietary models consistently fall short when faced with fine-grained linguistic restrictions in Italian and that stricter requirements tend to degrade output quality more severely. These findings motivate our decision to go beyond prompting and to opt for

different types of interventions.

Mechanistic interpretability and SAEs Beyond prompting, a more direct approach to output control is logit masking, i.e. forcing forbidden token probabilities to $-\infty$ at every decoding step, as implemented in frameworks such as that proposed by Willard and Louf [15] and its predecessors Hokamp and Liu [16]. At a deeper level, concept manipulation via internal model representations offers an alternative approach. Rather than suppressing surface tokens, it targets the concepts encoded in the model’s latent space using Sparse AutoEncoders (SAEs) [17]. They decompose dense LLM activations into sparse, monosemantic features [18] that can be ablated at inference time without modifying model weights. Furthermore, a feature’s interpretability does not guarantee its effectiveness as a steering target: features that cleanly represent a concept may have little causal influence on the model’s output – a gap whose limits we help characterise in the context of lexical avoidance.

3. Methodology

3.1. Data and models

The datasets consists of 194 italian Taboo cards manually transcribed from a physical version of the board game. Each example contains a target word and a list of forbidden words that cannot be used in the description.

We experiment with two open-weight models. `google/gemma-3-4b-it` [19] is a lightweight instruction-tuned model selected for its strong general performance and for the availability of pre-trained SAEs¹ compatible with our interpretability-based experimental condition. `openai/gpt-oss-20b` [21] is a Mixture-of-Experts reasoning model post-trained with chain-of-thought reinforcement learning, chosen to assess the role of explicit reasoning in constrained description generation, evaluated with and without reasoning. Generating an effective Taboo description requires implicitly planning ahead: the model must produce a description that is semantically informative while simultaneously avoiding a set of lexically related forbidden words. We therefore expect explicit reasoning to play a meaningful role in this task.

3.2. Forbidding experiments

We investigate three degrees of forbidding, each intervening at a different level of the model’s behaviour. In the first condition, models are *prompted* with the target word and the list of

¹`google/gemma-scope-2-4b-it` [20]



Figure 1: An example of a Taboo card. The word to be guessed is on top. The five words below are forbidden from appearing in the description.

forbidden words, analogously to how human players are instructed before their turn. The second condition moves the intervention downstream, *constraining the model during generation* so as to directly block forbidden tokens from being produced. The third condition operates at the level of internal representations, *manipulating the model’s latent space* to suppress the encoded concepts corresponding to the forbidden words.

3.2.1. Prompting

The prompt is constructed programmatically for each card, embedding the list of forbidden words and the target word directly into the user message. To ensure clean and comparable outputs, we additionally provide output instructions specifying that the model should return only the description, with no surrounding text, and within a limit of 30 words. The prompt explicitly instructs the model to avoid not only the forbidden words verbatim, but also any word derived from the same morphological root, in order to fully comply with the rules of the game. The full prompt structure is provided in Table ?? of Appendix A. Notably, the prompt does not explicitly instruct the model to avoid using the target word itself in the description, an error that would trivially make the model wrong at the game. We deemed it preferable to assess the extent to which this occurs empirically, and defer this analysis to Section 3.3. In Appendix A we additionally report results obtained with a relaxed prompt variant that omits the morphological constraint, discussing the impact of this simplification on model behaviour.

3.2.2. Generation-time constraints

A priori stems censorship Before generation begins, we compute a static set of banned token from the forbidden word list. For each forbidden word, we extract all stems and ban any vocabulary token whose surface form is a prefix of any such stem or lemma, or vice versa. The banned token are then masked by forcing their logits to $-\infty$ at every decoding step, making them impossible to sample regardless of context. A key design choice is the use of morphological normalisation rather than exact string matching: the censorship extends beyond the literal forbidden words to their inflected and derived forms. As a consequence, the model is forced to use synonyms or periphrases to express the forbidden elements. Complete freedom is left to the reasoning process.

Inference-time constrained generation A limitation of the *a-priori* approach is that it cannot condition the ban on generation context, occasionally suppressing tokens that would be appropriate in the description but happen to share a morphological root with a forbidden word. A practical example of its aggressiveness can be found in Appendix B. We address this by constraining the reasoning process dynamically. Rather than banning tokens upfront, we allow the model to generate freely and intervene only when a complete word is detected; its stem is then compared against the forbidden ones. If a match is found, generation backtracks to the first token of the offending word, which is added to a persistent position-level mask, and generation resumes from that position with the triggering token suppressed. Compared to the *a priori* approach, it operates on complete surface forms in place of subword fragments, avoiding spurious bans and higher computational overhead due to potential backtracing.